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Tapio Antila

DYNAMIC KELLY CRITERION -BASED PORTFOLIO LEVERAGE OPTIMIZATION

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TIIVISTELMÄ

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Kellyn kriteeri on teoreettisesti houkutteleva menetelmä pitkän aikavälin vaurauden maksimointiin sijoitusstrategioissa, mutta sen käytännön soveltaminen tuo mukanaan lukuisia haasteita. Tämä johtuu muun muassa mallin herkkyydestä parametrien arviointivirheille sekä mallin taustalla piilevistä epärealistisista oletuksista.

Tutkielma tarkastelee Kellyn teoriaa ja dynaamisen Kellyn soveltamista salkun viputason optimointiin käyttäen historiallista markkinadataa. Tavoitteena on selvittää, toimiiko Kellyn kriteeriin perustuva viputason optimointi teoriansa mukaisesti.

Työssä tarkastellaan Kellyn kriteerin historiaa, matemaattista mallia ja sen tulkintaa, oikean elämän sovelluksiin vaikuttavia rajoitteita sekä osittais-Kelly-strategiaa. Kellyn parametrien tulkinnan soveltamista pohditaan dynaamisen mallin tapauksessa, jonka jälkeen taustoitetaan tutkimuksen koeasetelma.

Tutkielmassa simuloidaan kolmen erilaisen Kelly-strategian avulla luodun portfolion kehitystä vuosina 1994–2024. Portfoliot koostuvat S&P 500 -indeksistä sekä korkopapereista. Sijoitusportfolion omaisuusluokkien suhde määritetään Kellyn kriteerin avulla käyttäen S&P 500 -indeksiä, Yhdysvaltojen kolmen kuukauden valtionvelkakirjojen korkotasoa sekä VIX-indeksiä.

Portfolioiden suoritusta analysoitaessa havaittiin, että vaikka Kelly-strategiat toimivat pääosin matemaattisen teorian ja aiemman kirjallisuuden mukaisesti, ne eivät tuottaneet tarkoituksensa mukaista maksimaalista portfolion kasvunopeutta. Tulokset tukevat aiempaa tutkimuskirjallisuutta Kellyn kriteerin ja sen käytännön sovelluksen haasteista.

Avainsanat: Kellyn kriteeri, portfolioteoria, velkavipu, dynaaminen Kelly

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1. Introduction

Portfolio optimization has long been a central theme in financial economics, with various strategies developed to maximize returns while managing risk. Among the more theoretically appealing approaches is the Kelly criterion, also known as the *Fortune's formula,* which originally formulated within the context of information theory. The criterion determines the optimal fraction of capital to allocate in a risky asset in order to maximize the long-term geometric growth rate of wealth (Thorp, 1962). While its mathematical foundations are well-established and its theoretical performance is impressive, the Kelly criterion is seldom applied in real-world portfolio management, at least not in its full form. This is largely due to practical concerns such as estimation errors in input parameters, unrealistic assumptions behind the formulas derivation, and behavioral deviations among investors.

These challenges, however, make the Kelly criterion all the more compelling to examine in realistic settings. Most existing research on the topic relies on simulations or stylized static variable assumptions that often diverge from actual market dynamics. The motivation for this thesis arises from the gap between theoretical models and real-world applicability: *Is a dynamically adjusted leverage strategy based on the Kelly formula the truly optimal growth rate strategy, as the theory suggests, when applied to real historical market data?*

This thesis investigates that question by implementing the Kelly criterion in a 31-year backtest from 1994 to 2024 with actual market data. Instead of relying on synthetic return distributions, the study uses the S&P 500 index as a risky asset, the VIX-index as a proxy for expected volatility, and the 3-month U.S. Treasury bill rate as the risk-free rate. The analysis includes three strategies: full Kelly, half Kelly, and double Kelly. These are compared against a standard S&P 500 index investment with no leverage, and a growth rate model is built for Kelly and fixed leverage portfolios.

Following this introduction, Chapter 2 provides the theoretical framework, presenting the origin, mathematics, and critiques of the Kelly Criterion. Chapter 3 describes the data and empirical methods used in portfolio construction and simulation. Chapter 4 presents and analyzes the results of the simulated portfolios, comparing Kelly strategies to conventional ones, and finally, Chapter 5 summarizes the main conclusions and discusses the practical implications and limitations of the findings.

2. Theoretical framework

2.1 Development and history of the Kelly criterion

In 1948, American mathematicians Claude E. Shannon and Warren Weaver introduced their paper "A mathematical theory of communication" on information theory, which deals with the quantitative measurement issues of information transfer. The key idea of their paper was to propose a theoretical upper limit to data transmission, depending on the entropy, channel capacity, redundancy and the average information content in data transmission. (Shannon, 1948). In 1956 one of Shannon colleagues, John L. Kelly Jr. applied Claude Shannon's information theory to gambling, formulating what is now known as the Kelly criterion. The criterion defines the optimal fraction of wealth to wager in order to maximize the long-term exponential growth rate of capital, based on the probabilities and payoffs of outcomes. (Kelly, 1956)

Edward Thorpe, a mathematics professor and a researcher in the field of gambling and investing expanded on the Kelly's criterion, addressing its application in optimizing bet size and leverage in stock investing. Thorp's model could calculate the Kelly criterion to bets with non-binary, continuous variable return rates, instead of just discrete bets. (Thorp, 1962). Thorp's work demonstrated that the Kelly criterion could be an effective tool in asset management, opening new possibilities for portfolio optimization.

2.2 The math behind the Kelly criterion

The mathematical proof for the Kelly criterion is quite extensive, and the original proofs can be found on Kelly's (1956) and Thorpe's (1962) papers, but the main points of the derivation will be represented here.

2.2.1 Derivation of the formula and fractional Kelly

In short, the goal of the Kelly criterion in investing is to maximize the expected growth rate of a long-term investment portfolio, which is mathematically equivalent to maximizing the expected value of the logarithm of wealth (Thorp, 1962). This can be presented as

$$E[\log(P_t)],$$

where P_t represents our portfolios value at time t.

In a simple case where we want to solve for the optimal distribution of assets in a portfolio consisting of only single asset and a risk-free rate, we model the price changes of the underlying asset as geometric Brownian motion

$$\frac{dS_t}{S_t} = \mu dt + \sigma dW_t,$$

where S_t is the price of the asset, μ is the drift of the underlying asset, σ is the volatility of the asset, and W_t is the Wiener process or Brownian motion that models the "randomness" of price fluctuations.

The price of the risk-free rate is modeled as

$$\frac{dB_t}{B_t} = r_f dt,$$

where r_f equals the risk-free rate.

By then constructing a portfolio consisting of these two assets, we can model the change of the portfolio's value

$$\frac{dP_t}{P_t} = f * \frac{dS_t}{S_t} + (1 - f) * \frac{dB_t}{B_t},$$

where f equals the portion of our portfolio invested in the underlying asset, and the fraction $(1 - f)$ is either the money invested or loaned at risk-free rate depending on whether $(1 - f) > 0$ or $(1 - f) < 0$.

By substituting equations for the change of bond and asset prices we get

$$\frac{dP_t}{P_t} = f * (\mu dt + \sigma dW_t) + (1 - f) * r_f dt$$

and

$$\frac{dP_t}{P_t} = (f\mu + (1 - f) * r_f)dt + (f\sigma)dW_t$$

Solving the stochastic differential equation and taking the logarithm of the solution shows

$$[\log(P_t)] = \log(P_0) + (f\mu + (1 - f) * r_f - \frac{1}{2}f^2\sigma^2)t + (f\sigma)W_t$$

This portrays the growth of the logarithmic value of our portfolio in relation to time.

Solving for the expected value of this function, the latter term $(f\sigma)W_t$ can be dropped off, since the expected value of the Wiener process is 0. After that we end up with the expected portfolio value

$$E\{[\log(P_t)]\} = \left(f\mu + (1 - f) * r_f - \frac{1}{2}f^2\sigma^2\right)t,$$

and because t is not dependent on other variables, we only have to maximize the function

$$g(f) = f\mu + (1 - f) * r_f - \frac{1}{2} f^2 \sigma^2$$

Maximizing this function to solve for optimal fraction can be done by differentiating the function with respect to f

$$g'(f) = \mu - r_f - f\sigma^2,$$

and solving for the roots of the derivative:

$$g'(f) = 0$$

and

$$\mu - r_f - f\sigma^2 = 0$$

From there, solving for f gives us the Kelly criterion

$$f = \frac{\mu - r_f}{\sigma^2}$$

This formula helps investors determine the proportion of their total wealth to allocate to a particular asset, balancing the expected return against the associated risk. The optimal investment fraction f^* increases with a higher expected drift μ, indicating that as the potential reward of an investment grows, a larger portion of capital should be allocated to it. Conversely, f^* decreases with higher risk-free rates r_f and greater volatility σ^2, reflecting a more cautious approach when safer investment alternatives are available or when the asset exhibits significant price fluctuations. (Thorp, 2008). This relationship underscores the trade-off between risk and return, a fundamental principle in financial decision-making.

2.2.2 Understanding the math

There are a few keynotes to be made from the derivation of the Kelly criterion, since there are a some assumptions that cannot be necessarily made when implementing the Kelly to real life scenarios.

Firstly the model used to showcase the change of prices in the risk free rate does not contain a parameter to model randomness. This means the model expects the bond price that represents the risk-free rate to have a standard price change in relation to time, which is not the case in real life (Cochrane, 2005, pp.349–382). The model also assumes that there is no limit to debt, and that the cost of debt and the interest on our money is always equal to the risk-free rate. For an asset manager that would manage a portfolio using the Kelly strategy and high levels of leverage, the default risk would be significant and the cost of borrowing would therefore most likely be higher than the risk free rate (Cochrane, 2005, pp.349–382).

The model also assumes the process to be continuous, meaning that the absolute amount of asset and risk-free rate bonds held in a portfolio need continuous rebalancing, which would be impossible, and would cause significant transaction costs for the portfolio manager. Even if the portions of the asset and bonds are only rebalanced daily or weekly, which may be enough in a long term, the transaction costs would still have an impact on the growth rate that is not considered in the derivation of Kelly.

2.3 Fractional Kelly

While the Kelly criterion provides a theoretically optimal strategy, its application in practical scenarios often necessitates modifications. One such modification is the fractional Kelly strategy, which involves investing a fraction less than the optimal Kelly amount. This approach is particularly useful in managing risk and mitigating

the effects of estimation errors in the parameters μ, r and σ^2. (MacLean, Ziemba & Blazenko, 1992, s.1562-1585)

2.3.1 The mathematical model for fractional Kelly

The mathematical formula for fractional Kelly is:

$$(2) \quad f = cf^*$$

where:

f is the amount of capital invested using the fractional Kelly strategy.
f^* is the optimal fraction of capital as determined by the standard Kelly formula.
c is the fraction of the optimal Kelly amount that you choose to invest, usually $0 < c < 1$

Fractional Kelly investing, often referred to as "fractional Kelly betting", adjusts the investment fraction to a level below f^*. For instance, a half-Kelly strategy would involve investing 50% of the Kelly optimal amount, where $c = 0{,}5$. This adjustment reduces the potential for extreme portfolio volatility and drawdowns, providing a more conservative approach while still benefiting from the growth optimization principles of the Kelly criterion.

Fractional Kelly strategies are often favored over the full Kelly criterion for several compelling reasons. Even though the full Kelly strategy gives the theoretical highest geometric excess growth rate, it's often considered too risky for investors (Thorp, 2011). There are numerous factors that should be considered when optimizing a portfolio that aren't considered in Kelly optimization, such as problems with the mathematical models assumptions, parameter estimations, handling management of drawdowns, the risk of black swan events, opportunity costs, risk tolerance, and the practical limitations of achieving long-term growth. (Thorp, 2011)

2.3.2 Sensitivity to parameter estimates

Unlike in information theory, where the formula was originally derived from, the inputs of the formula in investing cannot be expected to be constants, but approximations based on historic results and future financial forecasts. Historical performance may not accurately predict future returns since market conditions, economic environments, investor behavior and company fundamentals can change, leading to unreliable estimated parameters. Predictive models, while useful, are based on assumptions that might not fully capture market dynamics, resulting in inaccurate forecasts (Fama, 1970). For example, an unforeseen black swan event, such as Covid-19, can significantly alter the expected return and the volatility of the market. Additionally, investor behavior and market sentiment, influenced by factors such as herd behavior, overreaction to news, and irrational decision-making, can cause actual returns and volatility to deviate significantly from those predicted by financial forecasting. This phenomenon has been extensively studied in behavioral economics. Keynes (1936) introduced the concept of animal spirits to describe how emotions and instincts influence economic decision-making, and Shiller (2000) emphasized how psychological and social dynamics drive market fluctuations beyond what fundamentals would suggest. This means that the parameters of expected return and volatility both have additional underlying risks due to potential estimation errors, which isn't considered when calculating the Kelly criterion.

Fractional Kelly addresses this issue by reducing the bet size, which mitigates the impact of parameter misestimation. By betting a fraction of the Kelly amount, investors can buffer themselves against the inevitable uncertainties and inaccuracies in their parameter estimates, thereby achieving a more stable and sustainable growth trajectory (Thorp, 2008). Thorp (2008) also showed that by betting more than the optimal f^*, investor would receive more risk and less return. This is worse than under betting because, although betting less than the optimal fraction f^* also gives the investor lower returns, it also reduces the volatility of the investment. Betting a portion of $cf = f^*$ where $0 < c < 1$ provides the investor with

protection against the risk of overbetting, as it takes a more conservative approach to leverage.

2.3.3 High volatility, drawdowns and risk of ruin

The full Kelly criterion can recommend excessively large bets, which can lead to significant drawdowns if the bet loses. Such high volatility can be detrimental to investors and increase the risk of potential ruin. Behavioral factors, such as loss aversion and the inability to endure large drawdowns, further exacerbate the problem. Loss aversion, a concept from behavioral finance, indicates that investors experience the pain of losses more intensely than the pleasure of gains, making them prone to poor decision-making in the face of substantial drawdowns (Shiller, 2000). Moreover, the risk of ruin or severe drawdowns is significant due to the leverage used in Kelly investing. Unexpected black swan events or economic recessions, which result in prolonged negative returns, can cause catastrophic losses when employing the full Kelly strategy.

Fractional Kelly reduces the bet size, thus decreasing volatility and the risk of substantial drawdowns, making the investment strategy more palatable for most investors (Thorpe, 2008). By betting only a fraction of the Kelly amount, investors can avoid the severe consequences of large losses and maintain a more consistent and manageable level of portfolio volatility. This is showcased in a study conducted by Carta & Conversano (2020) where they simulated portfolios showcasing the potential losses after 100, 1 000, 10 000 and 40 000 individual trades done by betting according to different Kelly strategies.

The results after 10 000 trades showed that the portfolio that bet using the fractional Kelly approach, where the fraction $c = 0.5$, has only a 0.2% chance of losing more than half of the original capital, while the median return was 365%. In comparison, the full Kelly strategy had a 5.8% risk of more than losing half of the capital, with a median return of 490%. If the investor would have over bet, and used a Kelly fraction

double the optimal Kelly, with $c = 2$, the median return would have dropped to 60% with 36,5% of porfolios ending up with less than half of the original capital. (Carta & Conversano, 2020) This illustrates the significant downside of overbetting and supports Thorp's (2008) claim that under betting is better than overbetting.

2.3.4 Practical Implementation issues

For institutional investors, the foremost priority is often to maintain a specific level of risk, which can conflict with the aggressive optimization strategies suggested by the Kelly criterion. Institutional investors are typically bound by strict risk management frameworks and regulations that aim to protect the invested capital and ensure a stable return. Utilizing the Kelly criterion, which suggests allocating a larger portion of the portfolio to higher expected return assets, may result in a higher risk of significant short-term losses. (Carta & Conversano, 2020). This increased risk could lead to substantial capital withdrawals, undermining the institution's stability and reputation. Therefore, institutional investors might find the full implementation of the Kelly criterion impractical for their risk-averse investment strategies, but it is still utilized to some extent by hedge fund managers like Warren Buffet (Carta & Conversano, 2020).

While institutional investors may face significant constraints, private investors could potentially utilize the Kelly criterion for portfolio optimization. Retail investors might be willing to take more risk in the short-term and stomach more downside risk, knowing that they are maximizing wealth in the long-term (Kannadhasan, 2015). However, private investors must consider the high transaction costs that accompany frequent portfolio adjustments based on the Kelly criterion. Each transaction incurs fees, which can erode the potential gains from the optimized strategy. Additionally, private investors might not have access to leverage, or the same level of sophisticated financial tools and data as institutional investors, making it challenging to implement the Kelly criterion accurately and efficiently.

Another suboptimality in Kelly's criteria implementation is its inefficiency in short timeframes. The Kelly criterion is designed to maximize long-term growth, which means it assumes a sufficiently long investment horizon to smooth out volatility and capitalize on compounding returns. In a short timeframe, however, the Kelly criterion can lead to substantial potential losses, as the strategy might recommend high-risk positions that take time to recover from market downturns. Consequently, the criterion's effectiveness is diminished over shorter periods, increasing the short-term risk and making it less appealing for investors with limited time horizons. (Thorp, 2011). This is demonstrated in the study conducted by Carta & Conversano (2020), which found that in short-term scenarios with only 100 or 1,000 trades, the results did not align with the main property of the Kelly criterion, but with 40,000 trades, the full Kelly strategy was able to outperform all other investment strategies.

3. Data and methodology

While the Kelly criterions mathematical properties have been extensively explored, most existing studies focus on theoretical derivations or simulated environments that rely heavily on Monte Carlo methods. The studies based on actual historic data usually expect a static Kelly, meaning the expected volatility and drift stay constant. These approaches, while informative, don't necessarily showcase the problems or issues faced when applying the Kelly strategies in real life.

The purpose of this study is to examine how a portfolio that adjusts its leverage based on the Kelly criterion would have performed in practice in a 31-year period from the beginning of 1994 to the end of 2024. Instead of relying on artificial return distributions or stochastic models, this research constructs a real-world backtest using historical data from the S&P 500 index as the asset, the 3-month U.S. Treasury bill rate as a proxy for the risk-free rate bond, and the VIX index as a forward-looking measure of volatility.

Different Kelly strategy based portfolios and their performance will be simulated, and compared to both a traditional buy-and-hold investment in the S&P 500, and portfolios using fixed leverage ratios, to determine whether a Kelly formula based strategy would have given any edge to investors or not.

3.1. Framework for the research

This study simulates portfolios composed of the S&P 500 index and the 3-month U.S. Treasury bill, which serves as the proxy for the risk-free asset. Although the Kelly criterion is derived under the assumption of continuous-time portfolio optimization, it can be applied in discrete form by rebalancing the asset distributions on a daily basis, and by using daily estimates for the relevant parameters (Carta & Conversano, 2020).

The portfolio simulation is made between January 1 1994, and December 31, 2024, giving us a an estimated $31 * 252 = 7812$ individual trading days, that represent the number of Kelly bets in this time period.

Three Kelly strategies are simulated: full Kelly, fractional Kelly, and double Kelly. These portfolios are compared against both the no-leverage S&P 500 benchmark portfolio, or in other words the S&P 500 index, and fixed-leverage portfolios to assess relative performance in terms of return and risk.

3.2 Data and methodology

The simulation is built using publicly available historical market data retrieved using the quantmod package in R. The following are the indices and data used in the simulation.

S&P 500 Index: The index represents a broad measure of U.S. equity market performance and is used in our simulation as the primary risky asset. The data is taken from Yahoo Finance and the adjusted daily closing prices are used to compute both arithmetic and logarithmic returns.

VIX Index: The CBOE Volatility Index reflects the market's 30-day forward-looking implied volatility for the S&P 500 (Whaley, 2009). The data for adjusted daily closing price of the VIX index is sourced from Yahoo Finance.

3-Month Treasury Bill (DTB3) : The DTB3 represents the secondary market rate for 3-month Treasury bills. The daily annualized yields are retrieved from the Federal Reserve Economic Data (FRED) database. The annualized yield is then used to approximate the daily yield by converting the values to daily rate by dividing it by 252.

To ensure data continuity, all missing values in the S&P500, VIX and DTB3 series are linearly interpolated. All data series are aligned on a daily frequency and merged into a single dataset for analysis.

3.3 Calculating Kelly

Each of the parameters in the Kelly formula are calculated using the structured database depicted above. The parameters used to compute Kelly are only estimates, as the daily market volatility and returns are incredibly hard, if not impossible to be accurately predicted (Fama, 1965). The difficulties of the Kelly criterions applications are not only due to the difficulties caused by parameter predictions, but also due to the interpretation of the mathematics behind the Kelly formula.

One clear misunderstanding of the Kelly criterion and it's applications in the real world comes from the definition of the variable μ. A lot of the academic papers and research assumes the variable to be equal to the expected log return of the underlying asset. This is however not the case. When solving for the stochastic differential equation that describes the changes of a lognormal asset price, the solution would be

$$S_t = S_0 \exp\left((\mu - \frac{\sigma^2}{2})\right)t + \sigma W_t$$

From here we can see that the factor depicting the deterministic trend of the asset price, or the expected logarithmic return is

$$\mathrm{E}(\mathrm{R_a}) = \mu - \frac{\sigma^2}{2}$$

This connection is clear from the derivation of the formula, but it's applications to real-world scenarios seem unclear. In the stochastic differential equation this is

derived from, the drift μ and the σ percentage volatility are expected to be constants, which is not the case in real life. Changes of expected volatility in assets is a well-known phenomenon, and it can be predicted to some extent (Taylor, 2011, pp. 397–420).

Deriving an estimate for variance is straightforward, and can be done using the VIX index (Edwards & Preston, 2017). However, making accurate estimations on the mean of future daily log returns are incredibly difficult to be made, especially on short timeframes (Black, 1993). To make things even more difficult, there is very little academic literature discussing the relationship between the drift and expected logarithmic asset return. When implementing the Kelly criterion as a dynamic model, it seems unclear how the drift factor and expected asset log return should behave when expectations on volatility change. My interpretation is that the expected log asset return should not change under changes in volatility, since the changes in expected volatility would be immediately priced in, causing the asset price to change and therefore changing μ for the future while keeping $\mathrm{E}(\mathrm{R_a})$ a constant.

Therefore, in this research an estimate for μ is made by making estimates of the expected logarithmic asset return and implied variance

$$\mu = \mathrm{E}(\mathrm{R_a}) + \frac{\sigma^2}{2}$$

3.3.1 Defining μ

The variable μ is calculated for each day by making estimates for expected daily asset log return $\mathrm{R_{a,daily}}$ and variance σ^2_{daily}

$$\mu_{daily} = \mathrm{E}\left(\mathrm{R_{a,daily}}\right) + \frac{\sigma^2_{daily}}{2}$$

The estimation for $R_{a,daily}$ is made based on the historic data of the daily logarithmic returns of the S&P500. This is often used as an approximation in statistical finance (Campbell, Lo & MacKinlay, 1997). The estimated $R_{a,daily}$ is calculated as the average daily log returns from January 1st, 1900, up to the day preceding the current date. This long time window is chosen to minimize the momentum effects of rolling averages.

3.3.2 Defining σ^2

The predictions of the expected volatility for the next day are derived using the VIX index. The VIX index represents the annualized expected volatility of the S&P 500 index over the next 30 calendar days, meaning we can derive an estimate for the next day's VIX using this formula:

$$\sigma^2_{daily} = \frac{(VIX)^2}{2\,520\,000}$$

The derivation for this formula can be found in the appendices section.

3.3.2 Defining r_f

The risk free rate is estimated with

$$r_{f,daily} = \frac{r_{f,annualized}}{252},$$

where $r_{f,annualized}$ is the annualized yield of the 3-month treasury bond, and 252 is the approximated amount of trading days in a year.

3.4. Simulating and analyzing the portfolios

the Kelly portfolios are simulated by calculating the daily arithmetic profits of the strategies for each day and constructing an index using these daily returns. The arithmetic profits for each day are calculated using the formula:

$$portfolio_{rr,d} = cf * (S\&P\ 500)_{rr,d} - (cf - 1) * \left(r_{f,daily}\right),$$

where rr, d means realized daily returns.

The portfolios are analyzed by comparing the results to previous studies and the theorical framework. Analyzing metrics such as portfolio return, max drawdowns, realized compounding annual growth rate (CAGR) and annualized volatility are used. The realized growth rate of the different Kelly strategies are compared to different standard leveraged portfolios to try and determine whether or not the Kelly strategies managed bring optimal growth.

4. Results and analysis

This chapter presents and analyzes the empirical results of the simulated portfolios. Section 4.1 focuses on evaluating the performance and risk characteristics of the three different Kelly strategies, comparing them to each other, to the S&P 500 benchmark portfolio, and to findings from previous literature. Subsequently, Section 4.2 compares the realized growth rates of Kelly strategy - based portfolios to portfolios using fixed leverage, showing how the Kelly strategy fulfills its purpose of being the optimal growth model.

4.1 Performance metrics of the three Kelly portfolios

The following presents the performance of three different Kelly based portfolios and a benchmark portfolio from the beginning of 1994 to the end of 2024. The portfolio performances are evaluated using various metrics and compared with each other, recent literature, and the underlying mathematical framework.

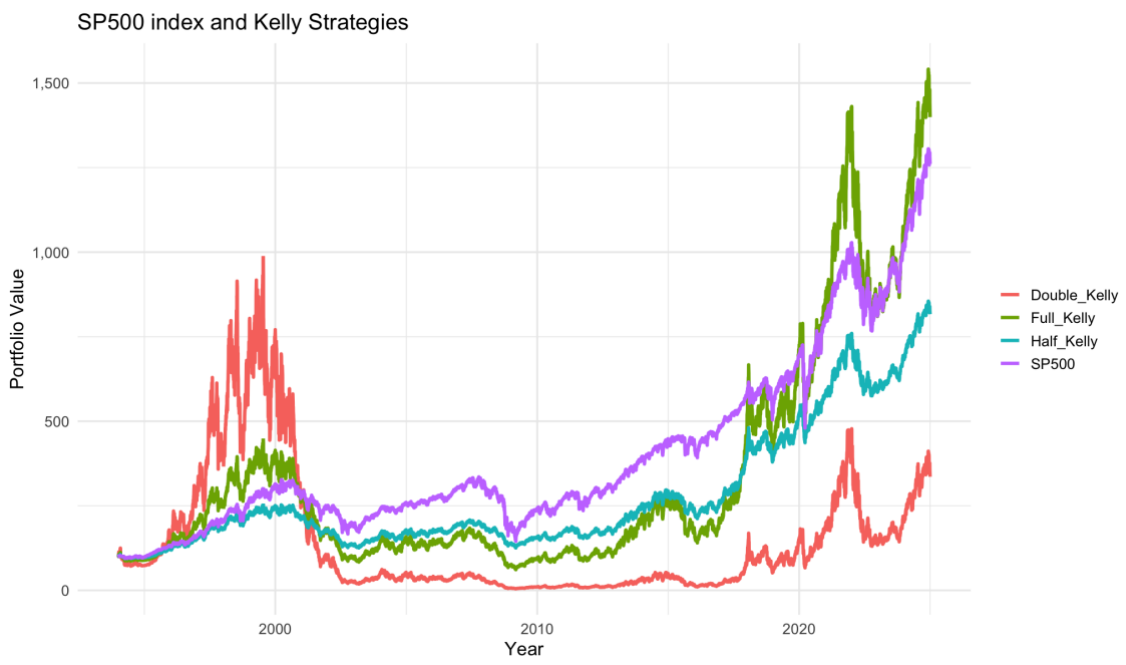

Figure 1: Kelly portfolios performance compared to benchmark portfolio, which is the S&P 500 index. Graph shows development of equity in portfolio, not accounting debt.

Portfolio	Min.Lev.	Max.Lev.	Avg.Lev.	Max.Drawdown	Annualized.Volatility	CAGR
Half kelly	0.053	3.844	1.200	51.10 %	16.18 %	7.02 %
Full kelly	0.106	7.688	2.402	86.40 %	32.36 %	8.91 %
Double kelly	0.213	15.376	4.803	99.54 %	64.27 %	4.04 %
S&P500	1.000	1.000	1.000	56.80 %	18.63 %	8.53 %

Table 1: Analytics and Performance metric of different portfolios from the whole period

The results of the simulated portfolios well align with the current literature and research regarding the Kelly criterion. As explained in *Understanding Kelly* (Thorp, 2011), the higher the Kelly fraction the higher the volatility. A increased fraction of wealth in risky asset increased the overall volatility of the portfolio as expected. Interesting note is, that even though the average leverage of the half Kelly portfolio is greater than the leverage of the S&P 500 benchmark portfolio, the annualized volatility decreased. This could be connected to the dynamic calculations of volatility in the simulation. Under volatile market conditions the VIX index value is greater which in our model would have increased σ^2_{daily}, reducing the value of optimal Kelly. Therefore leverage was decreased under volatile times, which could have led to annualized volatility being lower.

These findings are consistent with the Monte Carlo simulation study by Carta and Conversano (2020), which demonstrated that the full Kelly strategy yielded the highest median optimal growth rate, followed by the half Kelly and then the double Kelly strategies. Their results, based on a 10,000-trade simulation, are particularly comparable to our 7,812-trade simulation (252 trading days over 31 years), supporting the relative performance ranking observed in our analysis.

Carta and Conversano (2020) however found the highest mean from the double Kelly strategies when comparing these, but this is most likely caused by the outliers where double Kelly returned massive returns due to “bull market” scenarios. The

same phenomenon can be seen in our *Figure 1* plot. In the early bull market the double Kelly cumulated massive returns, but eventually almost went to ruin due to excessive leverage.

The realized CAGR was highest with the full Kelly strategy which was the only portfolio that outperformed the S&P 500 in this time period. The changes in leverage varied significantly over the time period, with max leverage being approximately $7{,}7$ with the full Kelly strategy and $15{,}4$ with the double Kelly. The full Kelly 7,7x leverage seems unrealistic, and even if the same portfolio could be replicated through options, the increased costs caused by high leverage, paired with the transaction costs not taken into account in our simulation, the slightly better CAGR may not seem appealing to an investor. Especially since the annualized volatility and max drawdowns in the full Kelly portfolio were significantly larger when compared to the S&P 500 benchmark portfolio.

4.2 Growth rate comparison to standard leverage portfolios

The following graphs showcase the daily realized growth rate of different Kelly based and fixed leverage portfolios, calculated using the formula

$$g = \left(\frac{W_n}{W_0}\right)^{\frac{1}{N}} - 1,$$

where W_n represents the portfolio's value at the end of the simulation, W_0 is the value of the portfolio in the beginning, and N is the amount of trading days.

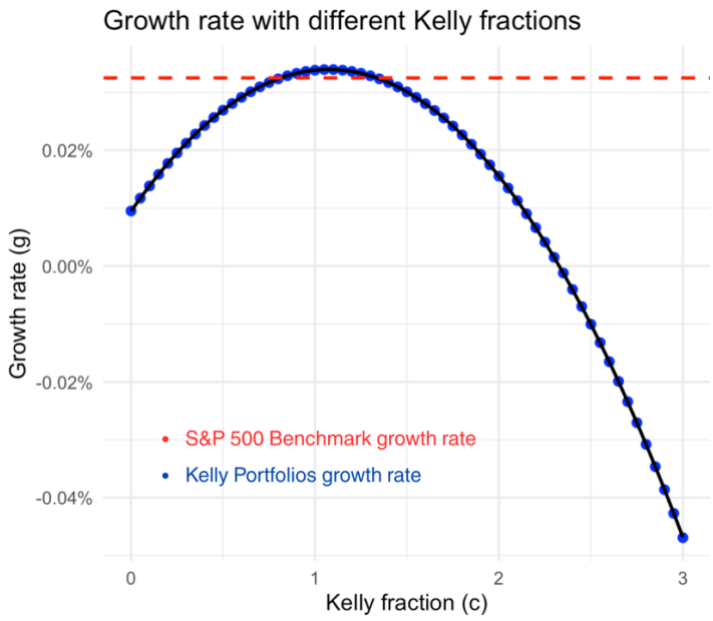

Figure 2: Realized growth rate of different Kelly fractions compared to the no-leverage benchmark S&P 500 index growth rate. NOTE: this chart differs from the usual Kelly growth rate chart, since the x-axis represents the Kelly fraction instead of leverage, due to dynamic Kelly modeling.

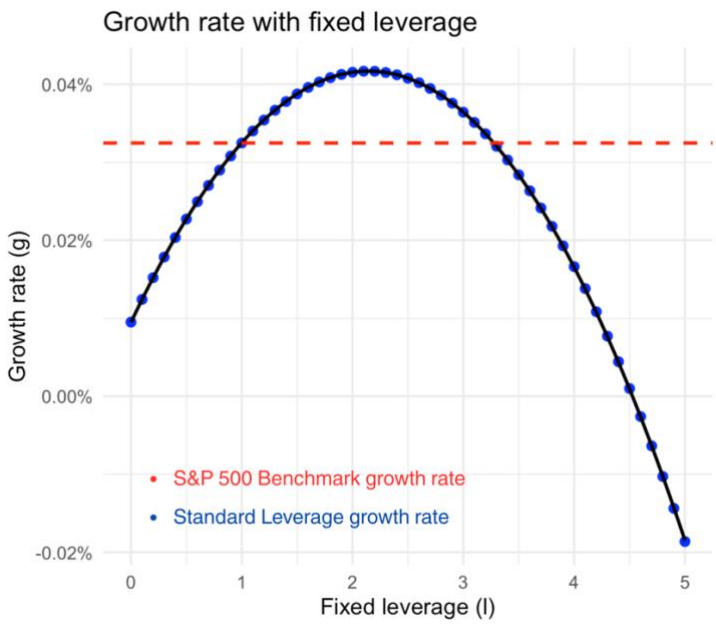

Figure 3: Realized growth rate of different fixed leverage portfolios compared to the no-leverage benchmark S&P 500 index growth rate. X-axis representing leverage.

The Kelly growth rate chart is very close to the theoretical growth rate model proposed by Thorp (2008) since the optimal Kelly growth rate in the 1994-01-01 to

2024-12-31 period would have been achieved with a Kelly fraction of approximately $c = 1{,}07$, compared to the theoretical optimal $c = 1{,}00$. However, the Kelly strategy did not show optimal growth when compared to the fixed leverage strategy. Kelly's criterion should give us the optimal growth rate out of all investment strategies (Thorp, 2011), but the fixed leverage strategy outperformed all the Kelly portfolios when the leverage was fixed at $1{,}10 < l < 3{,}20$.

This appears to contradict the core purpose of the Kelly criterion, since it should deliver the maximum growth rate for our portfolio when compared to any other strategies. This theoretically could be explained by the short time frame. In the Carta & Conversano (2020) study they found that not all simulations fulfilled the properties of the Kelly criterion every time with 10 000 trades. When the timeframe is short, it is possible that there are some strategies that manage to outperform the optimal Kelly strategy by sheer luck. Only when the timeframe was long enough with 40 000 trades, all of the Kelly conditions were met, and the optimal Kelly strategy outperformed all other strategies (Carta & Conversano, 2020).

However, the more likely explanation is, that the model did not outperform the standard leveraged portfolio due to parameter estimation errors. This theory is supported by the view in existing literature that the practical challenges of parameter estimation significantly undermine the Kelly criterion's theoretical promise of optimal growth (Thorp, 2011).

5. Conclusions

This bachelor's thesis examined the applicability of the Kelly criterion in real-world portfolio leverage optimization using historical market data from 1994 to 2024. Three simulated portfolios were constructed based on different implementations of the Kelly criterion: full Kelly, half Kelly, and double Kelly. Their performance was compared to a traditional buy-and-hold strategy in the S&P 500 index, after which a growth rate models were constructed to compare Kelly portfolios to portfolios with fixed leverage.

The results of the simulations align with existing literature and theoretical expectations. The full Kelly strategy yielded the highest compounding growth rate, but it also exhibited significantly higher volatility and deeper maximum drawdowns when compared to a standard S&P 500 benchmark portfolio. This supports the notion that while the Kelly criterion theoretically maximizes long-term geometric growth, its full implementation may be impractical for most investors due to its aggressive nature and sensitivity to estimation errors.

The half Kelly strategy demonstrated a more favorable risk-return profile. It offered lower volatility and drawdowns while still achieving decent growth rate compared to the S&P 500 benchmark portfolio. This highlights the value of fractional Kelly approaches in mitigating risks such as parameter misestimation and high volatility, but it also demonstrates why the fractional Kelly strategy is not the golden ticket to profits. It did not reach the same growth rate as a standard S&P 500 benchmark portfolio, meaning investors would have been better off investing all of their assets in an S&P 500 index fund.

The double Kelly strategy showcased the risks of overbetting, producing the worst drawdowns, despite strong early-period returns. This underlines the non-linear and asymmetric nature of risk in leveraged strategies: exceeding the optimal Kelly fraction not only increase risk but also reduced the returns.

When compared to fixed leverage strategies, the Kelly-based approaches did not outperform. In some cases, a well-chosen fixed leverage ratio provided better realized growth than the optimal Kelly strategy. This suggests that while the Kelly criterion provides a theoretical framework for optimal growth, its practical advantages may be limited without precise parameter estimation. This combined with the real life problems such as added transaction costs, stochasticity of bond price movements, and restricted asset liquidity, it seems unlikely any of the Kelly strategies would outperform a standard leveraged portfolio.

This research supports the theoretical foundations of the Kelly criterion but also emphasizes the challenges in real-world implementation. The foundation of the Kelly criterion and the theory of growth rate optimization can assist investors in conceptualizing portfolio construction from the perspective of maximizing growth, but applying this theory in real-world settings can prove challenging.

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Appendices

Derivation for the variance approximation:

VIX is represented as the annualized implied volatility of an asset

$$VIX = 100 * \sigma_{annual}$$

or

$$\frac{VIX}{100} = \sigma_{annual}$$

Daily volatility can be estimated with

$$\sigma_{daily} = \frac{\sigma_{annual}}{\sqrt{252}},$$

and variance can be estimated with

$$\sigma_{daily}^2 = \left(\frac{\sigma_{annual}}{\sqrt{252}}\right)^2$$

$$\sigma_{daily}^2 = \frac{\sigma_{annual}^2}{252}$$

$$\sigma_{daily}^2 = \frac{\left(\frac{VIX}{100}\right)^2}{252}$$

$$\sigma_{daily}^2 = \frac{VIX^2}{252 * 100^2}$$

$$\sigma_{daily}^2 = \frac{VIX^2}{2\ 520\ 000}$$

Simulation code (R):

```
#libraries
library(quantmod)
library(xts)
library(zoo)
library(PerformanceAnalytics)
library(ggplot2)
```

```
library(tidyr)

# Portfolio simulation

# S&P500
getSymbols("^GSPC", src = "yahoo", from = "1900-01-01", auto.assign = TRUE)
sp500_adjusted <- GSPC[, "GSPC.Adjusted"]
SP500_Arithmetic_Return <- dailyReturn(sp500_adjusted, type = "arithmetic")
SP500_Log_Return <- dailyReturn(sp500_adjusted, type = "log")

# Treasury
getSymbols("DTB3", src = "FRED")
rf <- DTB3 / 100
rf <- na.approx(rf)

# VIX
getSymbols("^VIX", src = "yahoo", from = "1990-01-01", auto.assign = TRUE)
vix_value <- VIX[, "VIX.Adjusted"]
vix_value <- na.approx(vix_value)

# daily rf and log rf
riskfree_daily <- rf / 252
rf_log_daily <- log(1 + rf) / 252

# variance from vix
vix_daily_variance <- (vix_value)^2 / 2520000
vix_daily_variance_lagged <- lag.xts(vix_daily_variance, k = 1) # lisätään lagi jotta
malli ei "ennusta" päivän VIX-closausarvoa

# merging series
merged_data <- merge(SP500_Arithmetic_Return, SP500_Log_Return, rf,
riskfree_daily, rf_log_daily, vix_daily_variance_lagged, join = "inner")
colnames(merged_data) <- c("SP500_Arithmetic_Return", "SP500_Log_Return",
"RiskFree_Rate", "RiskFree_Daily", "rf_log_daily", "VIX_Daily_Variance_Lagged")

# historic lagged average
returns_lagged_log <- lag.xts(merged_data$SP500_Log_Return, k = 1)
merged_data$Mean_Return_Log <- cumsum(na.fill(returns_lagged_log, 0)) /
cumsum(!is.na(returns_lagged_log))

# data timeframe
merged_data_2000_filled <- merged_data["1994-01-01/2024-12-31"]

# Kelly
merged_data_2000_filled$Optimal_Kelly <-
```

```
 (2*(merged_data_2000_filled$Mean_Return_Log-
merged_data_2000_filled$RiskFree_Daily)+merged_data_2000_filled$VIX_Daily_V
ariance_Lagged)/(2*merged_data_2000_filled$VIX_Daily_Variance_Lagged)

# Na values
merged_data_2000_filled$Optimal_Kelly[is.na(merged_data_2000_filled$Optimal_
Kelly)] <- 0

# Kelly variations
merged_data_2000_filled$Kelly_full      <- merged_data_2000_filled$Optimal_Kelly
merged_data_2000_filled$Kelly_fractional <-
merged_data_2000_filled$Optimal_Kelly * 0.5
merged_data_2000_filled$Kelly_double    <-
merged_data_2000_filled$Optimal_Kelly * 2

# renaming
riskfree_daily   <- merged_data_2000_filled$RiskFree_Daily
sp500_arth_return <- merged_data_2000_filled$SP500_Arithmetic_Return
kelly_frac       <- merged_data_2000_filled$Kelly_fractional
kelly_full       <- merged_data_2000_filled$Kelly_full
kelly_double     <- merged_data_2000_filled$Kelly_double

# profits
merged_data_2000_filled$Kelly_return_fractional <- (kelly_frac *
sp500_arth_return) - ((kelly_frac - 1) * riskfree_daily)
merged_data_2000_filled$Kelly_return_full      <- (kelly_full * sp500_arth_return) -
((kelly_full - 1) * riskfree_daily)
merged_data_2000_filled$Kelly_return_double    <- (kelly_double *
sp500_arth_return) - ((kelly_double - 1) * riskfree_daily)


# indices
sp500_index          <- cumprod(1 +
merged_data_2000_filled$SP500_Arithmetic_Return) * 100
kelly_index_fractional <- cumprod(1 +
merged_data_2000_filled$Kelly_return_fractional) * 100
kelly_index_full      <- cumprod(1 + merged_data_2000_filled$Kelly_return_full) *
100
kelly_index_double    <- cumprod(1 +
merged_data_2000_filled$Kelly_return_double) * 100

# xts
index_data <- merge(sp500_index, kelly_index_fractional, kelly_index_full,
kelly_index_double)
colnames(index_data) <- c("SP500", "Half_Kelly", "Full_Kelly", "Double_Kelly")
index_data <- na.omit(index_data)
```

 # Analysis

```
# leverage stats
leverage_stats <- data.frame(
  Strategy = c("Kelly_full", "Kelly_fractional", "Kelly_double"),
  Max_Leverage = c(
    max(merged_data_2000_filled$Kelly_full, na.rm = TRUE),
    max(merged_data_2000_filled$Kelly_fractional, na.rm = TRUE),
    max(merged_data_2000_filled$Kelly_double, na.rm = TRUE)
  ),
  Min_Leverage = c(
    min(merged_data_2000_filled$Kelly_full, na.rm = TRUE),
    min(merged_data_2000_filled$Kelly_fractional, na.rm = TRUE),
    min(merged_data_2000_filled$Kelly_double, na.rm = TRUE)
  ),
  Mean_Leverage = c(
    mean(merged_data_2000_filled$Kelly_full, na.rm = TRUE),
    mean(merged_data_2000_filled$Kelly_fractional, na.rm = TRUE),
    mean(merged_data_2000_filled$Kelly_double, na.rm = TRUE)
  )
)

# max drawdowns
sp500_cumindex          <- cumprod(1 +
merged_data_2000_filled$SP500_Arithmetic_Return)
kelly_cumindex_fractional <- cumprod(1 +
merged_data_2000_filled$Kelly_return_fractional)
kelly_cumindex_full      <- cumprod(1 + merged_data_2000_filled$Kelly_return_full)
kelly_cumindex_double    <- cumprod(1 +
merged_data_2000_filled$Kelly_return_double)

calc_max_drawdown <- function(index_series) {
  running_max <- cummax(index_series)
  drawdowns <- (index_series - running_max) / running_max
  return(min(drawdowns))
}

drawdown_stats <- data.frame(
  Strategy = c("SP500", "Fractional_Kelly", "Full_Kelly", "Double_Kelly"),
  Max_Drawdown_Percent = c(
    -calc_max_drawdown(sp500_cumindex) * 100,
    -calc_max_drawdown(kelly_cumindex_fractional) * 100,
    -calc_max_drawdown(kelly_cumindex_full) * 100,
    -calc_max_drawdown(kelly_cumindex_double) * 100
  )
)
```

```
# CAGR
get_cagr <- function(index_series) {
  index_series <- na.omit(index_series)
  start_value <- first(index_series)
  end_value   <- last(index_series)
  n_years    <- as.numeric(difftime(index(index_series)[length(index_series)],
                           index(index_series)[1],
                           units = "days")) / 365.25
  return ((end_value / start_value)^(1 / n_years) - 1)
}

cagr_fractional <- get_cagr(kelly_index_fractional)
cagr_full      <- get_cagr(kelly_index_full)
cagr_double    <- get_cagr(kelly_index_double)
cagr_sp500     <- get_cagr(sp500_index)

# Kelly growth rate

leverage_multipliers <- seq(0, 3, by = 0.05)

W0 <- 100
N <- nrow(merged_data_2000_filled)

growth_results <- data.frame(
  Kelly_Multiplier = leverage_multipliers,
  Growth_Rate = NA
)

for (i in seq_along(leverage_multipliers)) {
  multiplier <- leverage_multipliers[i]

  kelly_f <- merged_data_2000_filled$Optimal_Kelly * multiplier

  daily_return <- (kelly_f * merged_data_2000_filled$SP500_Arithmetic_Return) -
    ((kelly_f - 1) * merged_data_2000_filled$RiskFree_Daily)

  portfolio_values <- cumprod(1 + daily_return) * W0
  Wn <- tail(portfolio_values, 1)

  g <- (Wn / W0)^(1 / N) - 1
  growth_results$Growth_Rate[i] <- g
}

sp500_cumvalue <- cumprod(1 +
merged_data_2000_filled$SP500_Arithmetic_Return) * W0
sp500_Wn <- tail(sp500_cumvalue, 1)
```

```
sp500_growth_rate <- (sp500_Wn / W0)^(1 / N) - 1


# Fixed leverage growth rate

fixed_leverages <- seq(0, 5, by = 0.1)

W0 <- 100
N <- nrow(merged_data_2000_filled)

fixed_leverage_results <- data.frame(
  Leverage = fixed_leverages,
  Growth_Rate = NA
)

for (i in seq_along(fixed_leverages)) {
  lev <- fixed_leverages[i]

  daily_return <- (lev * merged_data_2000_filled$SP500_Arithmetic_Return) -
    ((lev - 1) * merged_data_2000_filled$RiskFree_Daily)

  portfolio_values <- cumprod(1 + daily_return) * W0
  Wn <- tail(portfolio_values, 1)

  g <- (Wn / W0)^(1 / N) - 1
  fixed_leverage_results$Growth_Rate[i] <- g
}

sp500_cumvalue <- cumprod(1 +
merged_data_2000_filled$SP500_Arithmetic_Return) * W0
sp500_Wn <- tail(sp500_cumvalue, 1)
sp500_growth_rate <- (sp500_Wn / W0)^(1 / N) - 1
```